Murat Ambarkutuk

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Professional Experience

R^2 Robotics - RANPAK **January 2025 – Present**
Principal Perception Engineer *In Person*

- Host engineering team meetings to coordinate cross-functional development efforts and align perception system objectives with product roadmap.
- Mentor perception team members on advanced computer vision techniques, deep learning optimization, and production deployment best practices.
- Architect the second-generation Decision Tower perception stack and features, including multi-camera 3D reconstruction, real-time object detection, TensorRT-optimized inference pipelines, CUDA-accelerated depth fusion, and TSDF voxel hashing for high-throughput packaging operations.

R^2 Robotics - RANPAK **November 2024 – January 2025**
Sr. Perception Engineer *In Person*

- Led development and deployment of production-grade computer vision systems for industrial automation, including multi-camera 3D reconstruction, real-time object detection, and TensorRT-optimized inference pipelines.
- Architected scalable perception frameworks with CUDA-accelerated depth fusion, TSDF voxel hashing, and neural scene representations for high-throughput packaging operations.
- Spearheaded model optimization initiatives including knowledge distillation from DINOv3 to YOLOv8, achieving real-time performance on embedded hardware while maintaining high accuracy for multi-class segmentation tasks.
- Developed complete perception stack for high-throughput packaging automation including multi-camera calibration, YOLO-based detection, and TensorRT-accelerated inference achieving real-time performance.
- Implemented production-grade TSDF fusion pipeline with voxel hashing for large-scale 3D mapping, supporting both depth-only and depth+color reconstruction with GPU-optimized memory access patterns.
- Built cross-platform TensorRT inference library (libinference) with dynamic shapes, multi-GPU support, CUDA stream parallelism, and unified configuration management for seamless deployment across development and production environments.

Baykar Defense (TR) - Atlas Imaging (US) **2018 – 2020**
Computer Vision Engineer developing orthorechtified maps with aerial photos taken by on autonomous UAVs. *Remote*

- Derived and actualize pose information (location+orientation) of an UAV and its components based on the distributed measurements of the Inertial Navigation System. These measurements are then used to in an iterative Computer Vision algorithm to project the corners of the images to the ground while considering the elevation changes in the scene.
- Developed a Sensor Fusion algorithm to correct results of the previous stages of the map-making algorithm with statistical representation.
- Implemented Test-driven Development paradigm in the dev-team. On that front, generated tests to check coverage, style requirements, and performance & accuracy of the developed algorithms.
- Architected the CI/CD pipeline to automate the aforementioned tests and generated automated reports from the results of the tests.

Education

PhD in Computer Engineering **2018 – 2024**
The Bradley Department of Electrical and Computer Engineering at Virginia Tech *Blacksburg, VA*

- Paul E. Torgersen Research Excellence Award
- The Outstanding Research Award and Fellowship from the Turkish Ministry of National Education

Dissertation: A Sensor Fusion Technique for Gait Analysis from Structural Vibration and Computer Vision

MS in Mechanical Engineering **2015 – 2018**
Mechanical Engineering at Virginia Tech *Blacksburg, VA*
Thesis: A Grid based Indoor Radiolocation Technique Based on Spatially Coherent Path Loss Model

BS in Mechatronics Engineering **2008 – 2013**
Mechatronics Engineering Department at Kocaeli University *Kocaeli, Turkey*
Thesis: Box Dimensioning System with Kinect Sensor

Technical & Language Skills

Languages: Python, C++, C, CUDA, MATLAB, HTML/CSS, JavaScript, PHP
ML/CV Frameworks: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, scikit-image, YOLO, DINOv2
3D Vision & Robotics: COLMAP, Multi-view Geometry, TSDF Fusion, 3D Gaussian Splatting, ROS
Deployment & Optimization: TensorRT, ONNX, CUDA Kernels, Quantization, Model Distillation
Tools & Technologies: Linux, Git, GitHub, GitLab, CI/CD, Docker, LaTeX
Language Proficiency: Turkish (Native), English (Proficient), German (Novice)

Projects

Decision Tower 2.0
R^2 Robotics/RANPAK **2025**
- Full perception stack rewrite for high-throughput packaging automation; multi-camera calibration, YOLO-based detection, and TensorRT-accelerated inference.

Quality Tower – Carton Graphic Validation & Hazard Label Inspection
R^2 Robotics/RANPAK **2025**
- Multi-class segmentation pipeline for on-line carton graphic validation and hazard-label inspection in industrial settings.

libinference – TensorRT Inference Library
R^2 Robotics/RANPAK **2024–2025**
- Header-only, cross-platform ONNX/TensorRT engine with dynamic shapes, multi-GPU support, CUDA stream parallelism, and unified logging/configuration.

Depth Regularizer
R^2 Robotics/RANPAK **2024–2025**
- Winsorized multi-view depth fusion with polynomial regression weighting and robust confidence filtering; production-grade C++/CUDA implementation.

TSDF Fusion with Voxel Hashing (CUDA)
R^2 Robotics/RANPAK **2024–2025**
- Real-time TSDF integration using voxel hashing; supports depth-only and depth+color fusion with GPU-optimized memory access for large-scale mapping.

Few-Shot Logo Classification with DINOv2
R^2 Robotics/RANPAK **2025**
- Prototypical-network architecture using a frozen DINOv2 backbone and raster logo support sets for robust industrial brand and logo recognition.

Knowledge Distillation – DINOv3 to YOLOv8
R^2 Robotics/RANPAK **2025**
- Distilling high-capacity DINOv3 features into lightweight YOLOv8 models to achieve real-time, high-accuracy detection and segmentation on embedded hardware.

Sensor Fusion for Occupant Localization in Smart Buildings
Advisors: P Plassmann, CF Jones | Related publications: [8, 10, 9, 2, 1] **2018–2024**
- Developed a robust algorithm that estimates heel-strike locations from the accelerometer measurements of a vibrating floor.
- Developed a Computer Vision-based algorithm to localize heel-strike locations from images taken by a stereoscopic camera.
- Developed a sensor fusion framework to incorporate the vibration- and vision-based heel-strike locations.
- Developed a combined simulation environment that can generate the vibration and visual information necessary for the validation of the algorithms developed.
- Quantified the efficacy and accuracy of implementations with simulations and controlled experiments.

Quantification of Collaboration and Communication among Students in a Smart Classroom
Advisors: T Baird, PA Tarazaga | News Article **2018**
- Designed a sensor-suite (frontend + backend), a multiple-view camera-based system, for a smart classroom to quantify the collaboration among students. The backend creates many functionalities of the suite, i.e., controlling the sensor-suite, scheduling auto-start/stop times as well as employing automated Computer Vision techniques on the collected data. The frontend, on the other hand, exposes the functionality of the backend to remotely control and monitor the system.

A Grid-based Indoor Radiolocation Technique based on Spatially Coherent Path Loss Model
Advisor: T Furukawa | Publications: [7] **2016–2017**
- Derived a probablistic framework of a passive radio-localization technique under multipath propagation that can determine the location of different entities, namely, occupants, robots, and invenvorty, in indoor enviroments.
- Developed the localization algorithm based on the probablistic framework.
- Validated the efficacy of the algorithm with controlled experiments whose results were presented at IEEE RAS conference.

Relevant Publications

[1] S. Alajlouni, **Ambarkutuk, Murat**, and P. Tarazaga. Towards the detection and localization of multiple occupant footsteps from vibroacoustic measurements. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 38th IMAC, A Conference and Exposition on Structural Dynamics 2020*, page 193. Springer, 2020.

[2] S. Alajlouni, V. V. S. Malladi, and **Ambarkutuk, Murat**. Investigation of using log-spectrum averaging (cepstral averaging) for blind reconstruction of an unknown impact input force. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 39th IMAC*. Springer, 2022.

[3] H. E. Güner, **Ambarkutuk, Murat**, L. Bilginer, and C. Oysu. Magnetic braking and speed stabilization. In *Proceedings of 18th International Conference*, pages 83–86. Proceedings of 18th International Conference. Mechanika 2013., 2013.

[4] H. E. Guner, O. Tekelioglu, **Ambarkutuk, Murat**, L. Bilginer, A. Aşkın, and S. Erkan. Distance mechatronic laboratory for sme's universities and vocational high schools in turkey. In *The 16th International Conference on Mechatronics, Mechatronica 2014*. Brno University of Technology, 2014.

[5] H. E. Guner, O. Tekelioglu, **Ambarkutuk, Murat**, L. Bilginer, A. Aşkın, and S. Erkan. Meslek yüksek okullari ve üniversiteler için internet tabanli mekatronik laboratuvari uygulamasi. In *3rd International Symposium on Innovative Technologies in Engineering and Science, 215*, 2015.

[6] H. Ocak, **Ambarkütük, Murat**, G. Küçükyıldız, and S. Karakaya. Image processing based package volume detection with kinect. In *2015 23nd Signal Processing and Communications Applications Conference (SIU)*, pages 515–518. IEEE, 2015.

[7] **Ambarkutuk, Murat** and T. Furukawa. A grid-based indoor radiolocation technique based on spatially coherent path loss model. In *2017 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, pages 220–226. IEEE, 2017.

[8] **Ambarkutuk, Murat** and P. E. Plassmann. Modeling and analysis of dispersive propagation of structural waves for vibro-localization. *Sensors*, 24(23):7744, 2024. doi: 10.3390/s24237744. URL https://doi.org/10.3390/s24237744.

[9] **Ambarkutuk, Murat** and P. Tarazaga. Uncertainty analysis of an occupant localization technique based on simulated structural vibrations. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 39th IMAC*. Springer, 2022.

[10] **Ambarkutuk, Murat**, S. Alajlouni, P. A. Tarazaga, and P. E. Plassmann. A multi-sensor stochastic energy-based vibro-localization technique with byzantine sensor elimination. *Sensors*, 23(23), 2023. ISSN 1424-8220. doi: 10.3390/s23239309. URL https://www.mdpi.com/1424-8220/23/23/9309.

Community Service & Leadership

Graduate Student Ambassador
The Bradley Department of Electrical and Computer Eng.

IEEE Robotics and Automation Society
Student Member & Reviewer

Society of Experimental Mechanics
Student Member

Turkish Student Assoc. at Virginia Tech
Secretary